

11. Analysis B: effective resolution in H, V, Z

Lateral resolution is read from the depth ESF across each knife edge. Two estimates are made and cross-checked: one from the slanted-edge fit, one from the robot-stepped poses. Z resolution is the step size detected with 50% probability under the same false-alarm rule used in D.

11.1 H and V resolution

1. Reference levels. For each edge and frame, fit planes to the front and back surfaces. Use only pixels farther than `BOUNDARY_BAND_HALF_WIDTH_PX` from the edge. These give Z_{front} and Z_{back} .
2. Normalized height. For each valid pixel, compute $h = (Z_{\text{back}} - Z) / (Z_{\text{back}} - Z_{\text{front}})$. Then $h = 1$ on the front surface and $h = 0$ on the back surface.
3. Signed distance. Compute each pixel center's signed distance s (px) to the true edge line, projected from the read-back pose carried through the registration and the as-built edge position, never from the commanded pose. s is positive on the front-surface side.
4. Robot-stepped ESF. Pool all frames and phase-jitter poses for that edge, Z , gap, and polarity. Bin by s at `BOUNDARY_BIN_WIDTH_PX`. ESF(s) is the mean h of the valid pixels in each bin. Keep the per-bin count of no-reads for Section 14.
5. Slanted-edge ESF. On the nominal-pose mean frame, locate the half-height crossing in each row (or column) and fit a line to the crossings. Then bin by distance to that fitted line. The robot-stepped and slanted-edge ESFs should agree to within one bin once aligned. If not, suspect the registration or a curved edge.
6. Metrics. The 10–90% rise distance is $s(h = 0.9) - s(h = 0.1)$, reported in px and in mm via $p(Z)$. The LSF is $d\text{ESF}/ds$, smoothed with a Hann window. The MTF is $|\text{FFT}(\text{LSF})|$ normalized to 1 at zero frequency. Report MTF50 in cycles/px. Rise distance is the primary figure, because a depth pipeline with interpolation is not a linear shift-invariant system, and an MTF assumes one.
7. Linearity test. Compare the ESFs at `GAP_SMALL_MM` and `GAP_LARGE_MM`. If they differ by more than 0.1 in h at any s , mark the result step-height dependent and quote resolution together with G .
8. Edge offset. The offset between the fitted edge line and the true one, $s_{50} = s(h = 0.5)$, is the edge bias. Section 14 uses it as the near/far preference.

9. Report for each Z, orientation (H or V), polarity, and gap. Confidence intervals come from bootstrapping over poses (BOOTSTRAP_RESAMPLES).

11.2 Z resolution

10. Sensed step. For each consecutive $A \rightarrow B$ visit pair, compute $\Delta = \text{mean}(Z \text{ at } B) - \text{mean}(Z \text{ at } A)$. Do this for patches of 1 px, 5×5 px, and 20×20 px, tiled over the ROI. The true step δ is the difference of the read-back target pose along the optical axis between the two visits; the registration's translation cancels in the difference, so only the robot's relative motion accuracy enters. Rungs smaller than ROBOT_REPEATABILITY_MM are flagged as unreliable truth and left out of the gain regression of Step 2. Also report the scatter of the read-back Z over the $A \rightarrow A$ pairs as the robot's own reported repeatability.
11. Gain and linearity. Regress the mean Δ on δ across the ladder. Report the slope (gain), the intercept, and the largest deviation from the line.
12. Null distribution and threshold. Differences between consecutive A visits ($A \rightarrow A$) form the no-step distribution for each patch size. Set τ at its $(1 - \text{DETECTION_FALSE_ALARM_TARGET})$ quantile.
13. Detection curve. A step counts as detected if $\Delta > \tau$ in the commanded direction. Fit the detection fraction against $\log \delta$ with the psychometric model of Section 13. Report $\delta_{50}(Z, \text{patch})$ with its confidence interval. For uncorrelated noise, δ_{50} shrinks as the square root of patch area. Shrinking more slowly than that means the noise is correlated over that scale, consistent with the correlation length from Section 10.
14. Ramp. For each ramp pose, subtract the fixed-pattern map of A at the same station, then average the depth along each image row within the plate's region of interest. The true depth of each row comes from the read-back pose of the tilted plate carried through the registration. Plot the row average against the true depth: the plateau widths give the measured quantum $\delta Z_q(Z)$; compare it with $q \cdot Z^2/k$ from Section 10. Plot single pixels the same way: a smooth row-average curve over a stepped single-pixel curve shows that the sensor's interpolation dithers the quantizer. Where the optional staircase was run, plot its sensed Z against the read-back Z and compare the two quanta.